RUINING YANG

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EDUCATION

Stony Brook University *Aug 2023 - Present*
Ph.D. - Computer Science
Advisor: Prof. Nick Nikiforakis

George Washington University *Aug 2021 - May 2023*
Master of Science - Computer Science
Advisor: Prof. Adam Aviv

Baylor University *Aug 2015 - May 2019*
Bachelor of Science - Computer Science
Minor: Mathematics

RESEARCH INTERESTS

Infrastructure-as-Code (IaC) security; software supply-chain security; credential and secret leakage; honeypot-based measurement of attacker behavior; LLM-based vulnerability detection.

PUBLICATIONS

[1] **Ruining Yang**, Nick Nikiforakis. *Measuring Security Hardening Adoption in Infrastructure as Code: An Empirical Study of Public Repositories.* International Conference on Privacy, Security and Trust (PST), 2026.

[2] **Ruining Yang**, Narong Chaiwut, Nick Nikiforakis. *Infrastructure as Compromise: Abusing Residual Trust in Infrastructure as Code Tools.* ACM Conference on Data and Application Security and Privacy (CODASPY), 2026.

[3] **Ruining Yang**. *User Perception of Login Notification Emails.* Master's Thesis, The George Washington University, 2023.

RESEARCH EXPERIENCE

Stony Brook University - Research Assistant Aug 2023 - Present

Software Supply-Chain Security in Infrastructure as Code

- **First study of residual trust in IaC**: playbooks fetch remote code and data at build time, so a source trusted today can later become unsafe when its domain expires or its host is compromised, giving attackers a supply-chain foothold that can silently reshape the deployed infrastructure.
- **463 hijackable domains and 10,400 misconfigured hosts across 247,131 repositories**: quantified the attack surface (46.9% of Ansible and 14.8% of Puppet modules allow remote references) and uncovered typosquatting plus a novel placeholder-domain attack vector.
- **Built DependoScope, a VS Code extension**: flags risky references (DNS, placeholder, typosquatting, service liveness) in real time; disclosed findings to affected maintainers.

Security-Hardening Adoption in Infrastructure as Code

- **Coined "ghost smells"**: security-hardening steps that should be present in IaC but are absent (missing firewall, unhardened SSH), inverting prior "security smell" work to flag missing protections rather than bad patterns.
- **A 32-rule hardening catalog and an LLM pipeline over 762,797 task instances**: distilled 50 public security guides and used GPT to analyze 256K Ansible and 34K Puppet repositories.
- **Only 10.81% of Ansible and 4.70% of Puppet repositories apply any hardening**: Linux dominates while web-server and multi-service hardening are nearly absent; released an `ansible-lint`-compatible linter.

Secret Leakage & Attacker Behavior on Code Platforms

- **300-repository controlled deployment**: measured secret leakage across 6 credential types, 5 leak formats (plaintext, Base64, concatenated, Ansible Vault with easy and hard passwords), and 10 hosting platforms.
- **Only 2 of 10 platforms sent leakage notifications**: GitHub and Hugging Face covered only a narrow set of plaintext and Base64 credentials, and no platform flagged concatenated or Ansible Vault formats.
- **Attackers attempted to use a leaked secret within 7 seconds**: over one month, tracked 2,126 authenticated credential-use attempts from 553 unique IPs across 34 repositories, with Base64 credentials receiving attempts as readily as plaintext.

George Washington University - Research Assistant Nov 2021 - May 2023

- Led a usable-security study on how users perceive login notification emails; designed the survey in Angular and recruited 100 participants on Prolific across seven randomized email conditions.
- Measured the real-world prevalence and design of login notification emails, and applied regression analysis to identify which factors (location, IP, browser, OS) help users judge whether a login was their own.
- Showed users weight delivery time (39.1%) and initial phrasing (40.9%) far more for login notifications than for regular emails, and that 86% view them as protective; distilled design guidance for account-security alerts.

Baylor Research Program - Research Assistant Jun 2018 - Aug 2018

- Researched Linux security; wrote lab procedures for Baylor's Competitive Cyber Security class.

TEACHING EXPERIENCE

Teaching Assistant, CSE 220 Systems Fundamentals I — Spring 2024
Teaching Assistant, CSE 220 Systems Fundamentals I — Fall 2023
Teaching Assistant, CSCI 2441W Database Systems and Team Projects — Spring 2023
Teaching Assistant, CSCI 2461 Computer Architecture I — Fall 2022

PROFESSIONAL SERVICE

Artifact Evaluation Committee, IEEE Symposium on Security and Privacy (S&P) — 2026
Artifact Evaluation Committee, Privacy Enhancing Technologies Symposium (PETS) — 2026

SKILLS

Programming	Python, C/C++, Java, SQL, JavaScript, TypeScript, PHP, HTML, LaTeX
Tools	Ansible, Puppet, Docker, Git, MySQL, Linux, Angular, Vue, Django, Flutter
Languages	English (Fluent), Mandarin (Fluent)

ACTIVITIES AND HONORS

Distinguished Artifact Reviewer, Privacy Enhancing Technologies Symposium (PETS) — 2026
Stony Brook University Chairman Fellowship — August 2023 - May 2025
Baylor Upsilon Pi Epsilon (Honor Society of Computer Science) — May 2019
Baylor Cyber Security Team — September 2017 - May 2019
Baylor Provost's Gold Scholarship — August 2015 - May 2019
Southwest Collegiate Cyber Defense Competition — March 2019